

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'The best type of laser with which to do spectroscopy over a
2. True or False: The correct answer to 'The rest mass of a particle with total energy 5.0 GeV and m
3. True or False: The correct answer to 'A rod measures 1.00 m in its rest system. How fast must a
4. True or False: The correct answer to 'A single-electron atom has the electron in the $l = 2$ state. The
5. True or False: The correct answer to 'Which of the following is true about any system that undergoes a reversible thermodynamic process?' is 'There are no changes in the internal energy of the system.'
6. True or False: The correct answer to 'The sign of the charge carriers in a doped semiconductor c
7. True or False: The correct answer to 'The energy required to remove both electrons from the heli
8. True or False: The correct answer to 'Electromagnetic radiation emitted from a nucleus is most lik
9. True or False: The correct answer to 'A refracting telescope consists of two converging lenses se
10. True or False: The correct answer to 'Which of the following is true about any system that undergoes a reversible thermodynamic process?' is 'There are no changes in the internal energy of the system.'

End of Paper